MCM/ICM Summary Sheet

Team Control Number: #2600000 **Problem Chosen:** C

Data-Driven Analytics and Predictive Modeling for Asian Giant Hornet Sightings:

Ranking Reports, Predicting Spread and Testing for Eradication

Abstract

Washington received 4,440 reports of possible Asian giant hornets between January and October 2020. Only 14 were confirmed, and 2,357 remained unresolved. We ranked unresolved reports with class-weighted ridge logistic regression using distance to the nearest known sighting, detection month, reporting lag, and two keyword flags. Across 20 repetitions of stratified five-fold validation, the model achieved average precision 0.4006, compared with 0.5037 for distance alone and 0.5380 for the random forest. The Brier score was 0.0054. The top 25 validation scores recovered 10 of 14 positives. At a relative distance-error standard deviation of 25%, average precision changed by -0.0051.

The Gaussian dispersal model placed 4 of 5 mainland clusters observed in 2020 inside the 95% region built from 2019 clusters. The cluster centroids shifted 10.3 km (95% bootstrap interval 2.6–22.2 km). We propose rescoring unresolved reports whenever a positive is confirmed. Under the conservative Poisson detection rate and a 90% prior probability that hornets remain, the modeled probability falls to 5% after 19.0 quiet active months, with field investigation of top-ranked reports also required.

Keywords: Asian giant hornet · Penalized logistic regression · Probability calibration · Dispersal kernel · Poisson detection · Eradication criterion

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1 Introduction

1.1 Dataset Overview & Problem Background

In September 2019, a nest of *Vespa mandarinia*, the Asian giant hornet, was found on Vancouver Island, British Columbia. Since then, several confirmed sightings have occurred in neighboring Washington State. The hornet is the largest in the world and a predator of European honeybees: a small group can destroy a whole honeybee colony in a short time. To track the pest, the State of Washington asks the public to report sightings. However, most reports turn out to be other insects, and the state has limited resources to follow up on each one.

Our goal is to use the public reports to build a model that estimates how likely each report is to be a true Asian giant hornet sighting. The state can then use this ranking to decide which reports to investigate first. We also use the data to discuss whether the spread of the hornet can be predicted, how the model should be updated as new reports arrive, and what evidence would show that the pest has been eradicated.

The dataset contains $N = 4{,}440$ public reports with $D = 8$ columns: a unique ID, detection date, reporter notes, lab status, lab comments, submission date, latitude, and longitude. Reports were submitted between January and October 2020. The data is almost complete, with only 2 missing values (both in Lab Comments), but some detection dates are invalid, such as years before 1900. The main challenge is extreme class imbalance. Only 14 reports are confirmed hornets (Positive ID), compared with 2,069 Negative IDs, 2,342 Unverified reports, and 15 Unprocessed reports.

1.2 Data Analytics Workflow

Our analysis follows five phases:

1. **Phase I (Data Cleaning):** Check coordinates and dates for invalid values, and label the reports. Positive IDs are labeled 1 and Negative IDs are labeled 0. Unverified and Unprocessed reports are kept unlabeled.

2. **Phase II (Feature Engineering):** Build a small set of features from the raw data: distance to the nearest other confirmed sighting, month of detection, days between detection and submission, keyword flags from the notes, and whether the report included a photo or video.

3. **Phase III (Predictive Modeling):** Train a classification model on the labeled reports and use cross-validation to tune it and measure its performance. Because there are only 14 positives, we focus on precision, recall, and F1-score rather than accuracy.

4. **Phase IV (Interpretation & Prioritization):** Identify which features drive the predictions, test how stable the model is when noise is added to the inputs, and use the predicted probabilities to rank unverified reports for investigation.

5. **Phase V (Spread, Updating & Eradication):** Predict where new nests can appear, set a schedule for updating the model, and derive a rule for declaring eradication.

2 Data Preprocessing & Feature Engineering

2.1 Data Cleaning & Imputation (See Appendix)

To prepare the Asian Giant Hornet sighting dataset for predictive modeling, we first identify missing values, duplicate records, and extreme outliers. Continuous features are standardized using Z-score normalization:

$$x_{\text{scaled}} = \frac{x - \mu}{\sigma} \tag{1}$$

x is the original feature value, and μ and σ are its mean and standard deviation on the training folds only.

$$\hat{x}_{ij} = \begin{cases} x_{ij}, & \text{if observed} \\ \text{median}(X_j^{\text{obs}}), & \text{if missing} \end{cases} \tag{2}$$

x_{ij} represents the value of feature j for sighting i, and X_j^{obs} represents the observed values of feature j in the training folds, so the median is recomputed inside every fold.

Fourteen detection dates are unusable (three blank, eleven before 1900). Thirteen belong to unverified reports and one to a Negative ID dated 1/21/1200, so no positive is affected; these reports take their month from the submission date and treat the lag as missing, which keeps all 2,083 labeled reports. Eighteen reports list a detection after the submission, and their lag is set to zero.

In addition, invalid geographic coordinates and inconsistent detection dates are flagged for review. Confirmed positive sightings are assigned a binary label of 1, while confirmed negative sightings are assigned 0. Unverified and unprocessed reports remain unlabeled to avoid introducing false classifications. Geographic data cleaning was performed by validating the latitude and longitude coordinates associated with each reported sighting. Inside cross-validation every quantity estimated from data comes from the training folds alone, including the distance to the nearest confirmed sighting: a held-out report is measured only against training-fold positives (Section 4).

2.2 Feature Engineering & Dimension Reduction

To help the model separate real hornet sightings from the mistaken ones, we built several features from the raw report data. We calculated the distance in kilometers from each report to the nearest confirmed hornet sighting since the confirmed hornets were concentrated near the Canadian border, and we use its logarithm because these distances run from a few meters to almost 500 km. For this calculation each report was compared only against other confirmed sightings so a confirmed report was never measured against itself. We extracted the month of each detection to capture the seasonal patterns in hornet activity, encoded it as a sine and cosine pair so that December sits next to January, and we calculated the number of days between detection and submission. For the reporter's notes, we created yes/no flags for the word "dead" and for words showing that a specimen was kept ("specimen", "collected", "caught", "killed", "captured"). We also built a yes/no flag for an attached photo or video from the image file list but left it out: a file is attached to 98.7% of negatives, 78.6% of positives and 2.9% of unverified reports, so the flag records whether the lab could process

a report, not whether it shows a hornet. The keyword "photo" appears in none of the 14 positive notes and was dropped too. Because only 14 reports had confirmed hornets we kept the number of features small, six in total, to avoid overfitting.

Table 1: Data Preprocessing and Feature Pipeline Summary

Raw Feature	**Transformation Applied**	**Engineered Feature**	**Rationale**
Latitude, Longitude	Haversine distance d_i to nearest other confirmed sighting (training-fold positives only), then $\ln(1+d_i)$, z-score	$\tilde{d}_i$	Confirmed hornets cluster near the Canadian border
Detection Date	Month m_i (submission month if the date is invalid), then $\sin(2\pi m_i/12)$ and $\cos(2\pi m_i/12)$, z-score	s_i, c_i	Captures seasonal hornet activity, with December next to January
Detection, Submission Dates	Days between the dates, $\max(0,\cdot)$, then $\ln(1+\cdot)$, training-fold median if missing, z-score	ℓ_i	Faster reports may be more reliable
Notes	Contains "dead"	k_i^{dead} (0/1)	A dead insect can be examined by the lab
Notes	Contains "specimen", "collected", "caught", "killed" or "captured"	k_i^{spec} (0/1)	A kept specimen allows lab confirmation
Image file list	Any photo or video attached	Not used	Attached to 98.7% of negatives but 2.9% of unverified reports

3 Assumptions and Notation

Table 2: Notation used throughout the paper

Symbol	**Meaning**	**Symbol**	**Meaning**
i	Report index, $1,\ldots,4440$	π	Base rate, $N_1/\lvert\mathcal{L}\rvert = 0.0067$
y_i	Lab label: 1 Positive, 0 Negative ID	$R,\ \sigma_D$	Queen range (30 km); kernel scale
$\mathcal{L}$	Labeled set: $N_1 = 14$, $N_0 = 2069$	$\mathbf{c}_j,\ n$	Nest cluster j; clusters per season
$\mathcal{U}$	Unverified or unprocessed (2357)	$\mu,\ \mu_L$	Clusters per active month; lower bound
$\mathcal{P}$	Confirmed positive reports	ρ_0	Prior probability population present
(φ_i, λ_i)	Latitude, longitude (radians)	γ	Tolerated eradication error
$\mathbf{x}_i,\ p_i$	Features; probability of true sighting	$T,\ T^*$	Quiet active months; months required

Of the 14 positives, one is on Vancouver Island near Nanaimo; we call the other 13 the *mainland positives*. They lie around Blaine, Washington, and two of them (13 November 2019 and 15 May 2020) are just north of the border in British Columbia. Each assumption below is paired with the check we run on it.

1. **Lab labels are correct.** Unverified reports are not assumed negative, because no determination was made. *Check:* the Notes or Lab Comments of 9 of the 14 positives cite a specimen, a captured hornet or an image; the other 5 give no evidence type, and we accept the lab's determination for them.

2. **Labeled and unlabeled reports come from the same population of reporters.** *Check:* two-sample Kolmogorov–Smirnov (KS) tests of each continuous feature between $\mathcal{L}$ and $\mathcal{U}$ gave the following results: log distance ($D = 0.061, p = 0.00054$), sine of month ($D = 0.180, p = 7.7 \times 10^{-32}$), cosine of month ($D = 0.078, p = 2.6 \times 10^{-6}$), log lag ($D = 0.174, p = 1.2 \times 10^{-29}$). Missing lag values were omitted. The distance comparison is descriptive because its positive reference set comes from the labeled group. The season and lag results suggest that scores learned on verified reports may transfer imperfectly to unresolved reports.
3. **Coordinates are accurate to within a few kilometers.** *Check:* the noise experiment of Section 4.6.
4. **Queens disperse in a random direction, one generation per year.** *Check:* Section 5 tests the 2019-based prediction region on the 2020 clusters and tests the one outlying positive.
5. **Reports from one nest are one event.** Several 2020 positives lie within a few hundred meters of each other, most likely one nest, so Models II and IV use *nest clusters*: same-year positives linked by distances of at most 1 km. *Check:* the 2019 to 2020 centroid shift is 10.2, 10.2, 10.3 and 12.7 km at report level and at 0.5, 1 and 2 km linkage; only the 2 km rule, which merges the four 2019 positives (1.4 to 10.2 km apart) into two clusters, changes it noticeably.
6. **While a population is present, positive clusters arrive as a Poisson process at a rate of at least μ_L per *active month*** (May to November, between queen emergence and overwintering [4, 5]). *Check:* the sample variance-to-mean ratio was 0.68 for the 2020 monthly counts; 12 of the 13 mainland positives were detected from May to November, the exception being one dead hornet reported on 8 December 2019. The rate for a small remnant cannot be tested until it is gone, so Model IV also requires the calculated quiet period and searches of top-ranked reports.

4 Model I: Probability That a Report Is a True Sighting

The problem asks for the likelihood that a report is a mistaken classification. We model the probability p_i that report i is a true Asian giant hornet, so the probability of a mistaken classification is $1 - p_i$. Notation follows Table 2.

4.1 Feature Definitions

Distance to the nearest confirmed sighting. The great-circle distance in kilometers between reports i and j uses the haversine formula with Earth radius $r_E = 6371$ km, and d_i is the distance to the nearest positive in a reference set $\mathcal{P}_{\text{ref}}$:

$$h(i,j) = 2r_E \arcsin\sqrt{\sin^2\!\left(\tfrac{\varphi_j-\varphi_i}{2}\right) + \cos\varphi_i \cos\varphi_j \sin^2\!\left(\tfrac{\lambda_j-\lambda_i}{2}\right)}, \qquad d_i = \min_{j\in\mathcal{P}_{\text{ref}},\, j\neq i} h(i,j). \quad (3)$$

Inside cross-validation, $\mathcal{P}_{\text{ref}}$ holds the positives of the *training folds only*: three positives from the September 2020 nest site lie within 8 m of one another, so two of them sharing a test

fold would otherwise find each other and be scored correctly for the wrong reason. Distances run from 5 m to 484 km, so the model uses $\tilde{d}_i = \ln(1 + d_i)$.

Season and reporting lag. The month m_i (from the submission date when the detection date is invalid) is placed on a circle, $s_i = \sin(2\pi m_i/12)$ and $c_i = \cos(2\pi m_i/12)$, so December sits next to January. The lag is $\ell_i = \ln\big(1 + \max(0,\, t_i^{\text{sub}} - t_i^{\text{det}})\big)$ in days, with the training-fold median used when t_i^{det} is invalid.

Keyword flags. $k_i^{\text{dead}} = 1$ when the notes contain "dead" (2 of 14 positive notes against 7.7% of negative notes), and $k_i^{\text{spec}} = 1$ when they contain "specimen", "collected", "caught", "killed" or "captured" (4 of 14 against 9.7%). Lab Comments and notes containing "WSDA" (3 of 14 positives, no negatives) are excluded because staff write them after the identification; Section 2 covers the media flag.
The feature vector is $\mathbf{x}_i = (\tilde{d}_i, s_i, c_i, \ell_i, k_i^{\text{dead}}, k_i^{\text{spec}})$; the four continuous features are standardized with training-fold means and standard deviations, and the flags stay 0/1.

4.2 Primary Model: Class-Weighted Penalized Logistic Regression

The model is $p_i = \sigma(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x}_i)$ with $\sigma(z) = 1/(1+e^{-z})$. Predicting "not a hornet" everywhere already scores 99.3% accuracy, and an unweighted fit stays near that answer. We weight the classes so both contribute equally to the loss, and add a ridge penalty because 14 positives give only $14/6 \approx 2.3$ events per coefficient:

$$\min_{\beta_0,\boldsymbol{\beta}} \; -\sum_{i\in\mathcal{L}} w_{y_i}\Big[y_i \ln p_i + (1-y_i)\ln(1-p_i)\Big] + \alpha\|\boldsymbol{\beta}\|_2^2, \qquad w_1 = \frac{|\mathcal{L}|}{2N_1}, \quad w_0 = \frac{|\mathcal{L}|}{2N_0}. \tag{4}$$

Numerically, $w_1 = 2083/28 = 74.4$ and $w_0 = 2083/4138 = 0.503$. The distance-only baseline uses $\alpha = 1$. The penalty α is chosen from $\{10^{-3}, 10^{-2}, \ldots, 10^{2}\}$ by an inner cross-validation that maximizes average precision, giving $\alpha = 100$ for the full-data fit.

Calibration back to the true base rate. Class weighting makes the model act as if half of all reports were positive, which inflates every probability without changing the ranking. A change in class prior multiplies the posterior odds by the ratio of prior odds, so we shift the log-odds:

$$\operatorname{logit} p_i^{\text{cal}} = \operatorname{logit} p_i^{\text{w}} + \ln\frac{\pi}{1-\pi} - \ln\frac{0.5}{0.5} = \operatorname{logit} p_i^{\text{w}} + \ln\frac{\pi}{1-\pi}, \tag{5}$$

where π is the base rate of the fitting data (training folds, or all of $\mathcal{L}$). The shift is exact for a correctly specified model and approximate under the penalty; the Brier score checks it. *Worked example.* If $p^{\text{w}} = 0.80$, then $\operatorname{logit}(0.80) = \ln 4 = 1.386$. With $\ln(14/2069) = -4.996$, the calibrated log-odds is $1.386 - 4.996 = -3.610$, so $p^{\text{cal}} = e^{-3.610}/(1 + e^{-3.610}) = 0.026$. A score that looks like 80% is a 2.6% chance, about four times the base rate of 0.67%.

4.3 Comparison Models

The **distance-only rule** is a logistic regression on $\tilde{d}_i$ alone with the same weights and calibration; it ranks reports exactly as $-d_i$ does, and if the full model cannot beat it, the other five features add nothing. The **random forest** has 300 trees, maximum depth 3, at least 5 reports per leaf and balanced class weights, since 14 positives cannot support deep trees; its probabilities are clipped to $[10^{-6}, 1-10^{-6}]$ and calibrated with Equation (5).

4.4 Validation Design and Metrics

We use stratified 5-fold cross-validation repeated $Q = 20$ times (2 or 3 positives per test fold), recomputing Equation (3), imputation, standardization and α inside each training fold. Each labeled report receives 20 out-of-fold probabilities; their mean is its score $\bar{p}_i$, and it counts as flagged when $p_i^{\text{cal}} \geq \tau$ in at least 10 of the 20 repeats, where τ maximizes F_2 on inner out-of-fold training predictions for each model. With true positives TP, false positives FP and false negatives FN, precision and recall are

$$\text{Prec} = \frac{TP}{TP+FP}, \qquad \text{Rec} = \frac{TP}{TP+FN}, \qquad F_\beta = (1+\beta^2)\,\frac{\text{Prec}\cdot\text{Rec}}{\beta^2\,\text{Prec}+\text{Rec}}. \tag{6}$$

We set $\beta = 2$, which weights recall four times as heavily as precision in the harmonic mean, because a missed nest can destroy many honeybee colonies while a wasted visit costs one trip. Average precision (AP), $\text{AP} = \sum_k (\text{Rec}_k - \text{Rec}_{k-1})\,\text{Prec}_k$, is the headline number when positives are rare; a random ranking scores about $\pi = 0.0067$. The Brier score $\text{BS} = \frac{1}{|\mathcal{L}|}\sum_i (\bar{p}_i - y_i)^2$ measures goodness of fit against the reference $\pi(1-\pi) = 0.0067$ of always predicting π. Recall can only equal $k/14$ for $k = 0, \ldots, 14$ (for example $12/14 = 0.857$), and every recall we report is one of these values.

4.5 Interval Estimates

Each metric gets a 95% percentile interval from $B = 2000$ stratified bootstrap resamples of the report-level out-of-fold results (positives and negatives resampled separately). Each coefficient is refit on $B = 2000$ stratified bootstrap resamples of $\mathcal{L}$ at the selected α, giving the odds ratio e^{β_j} and its percentile interval: the change in the odds of a true sighting per standard deviation for a continuous feature, or flag present versus absent. The sine and cosine terms are read together as one seasonal curve.

4.6 Robustness to Location Error

Public coordinates may be approximate, so we perturb the test-fold distances while training data stay clean:

$$d_i^{\text{noisy}} = d_i \max(0,\, 1+\varepsilon_i), \qquad \varepsilon_i \sim \mathcal{N}(0, \eta^2), \qquad \eta \in \{0.05, 0.10, 0.25\}. \tag{7}$$

Here η is the relative error in distance; each level averages the change in AP from $\eta = 0$ over 100 draws.

4.7 Prioritizing Reports for Investigation

Every report in $\mathcal{U}$ receives p_i^{cal} from the model trained on all of $\mathcal{L}$, with all 14 positives in $\mathcal{P}_{\text{ref}}$. **Capacity rule.** With capacity for K visits, the state investigates the K highest scores and expects to find $E[\text{positives found}] = \sum_{i \in \text{top } K} p_i^{\text{cal}}$ true hornets. Recall at K is the share of the 14 positives that rank in the top K of the out-of-fold scores $\bar{p}_i$ (2,083 labeled reports, close in size to the 2,357 in $\mathcal{U}$). **Cost rule.** If a visit costs C and a missed hornet costs L, report i is worth a visit when

$$p_i^{\text{cal}} L \geq C \quad \Longleftrightarrow \quad p_i^{\text{cal}} \geq \frac{C}{L}. \tag{8}$$

Worked example. If a missed nest is judged 100 times as costly as a visit, every report with $p^{\text{cal}} \geq 0.01$ is investigated, and the report from the calibration example ($p^{\text{cal}} = 0.026$) clears this bar.

4.8 Results

Interpretation. Weighted logistic regression achieved AP 0.4006 [0.2339, 0.6969], compared with 0.5037 [0.2602, 0.7384] for distance alone and 0.5380 [0.2924, 0.7758] for the random forest. Adding the other five features changed AP by -0.1031 relative to distance alone. The fold-specific F_2 thresholds flagged 10 of 14 positives with precision 0.435. The Brier score was 0.0054 [0.0045, 0.0064], compared with 0.0067 for the base-rate prediction. The out-of-fold probabilities summed to 7.75 against 14 observed positives. Log distance had the largest absolute coefficient, with odds ratio 0.368 [0.322, 0.436] (Table 4). Intervals containing 1 (sine of month, cosine of month, dead keyword, specimen keyword) leave those directions uncertain. At a relative distance-error standard deviation of 25%, AP changed by -0.0051. The summed estimated probability for the top 25 unresolved reports was 5.793 (Table 6).

Table 3: Out-of-fold performance of Model I and its comparison models (repeated stratified 5-fold cross-validation, 5×20). Intervals are 95% stratified bootstrap percentiles.

Model	Precision	Recall	F_2	Average Precision	Brier Score
Distance-only rule	0.5000 [0.2857, 0.7500]	0.5000 [0.2143, 0.7857]	0.5000 [0.2344, 0.7237]	0.5037 [0.2602, 0.7384]	0.0048 [0.0036, 0.0059]
Random forest	0.2963 [0.1714, 0.4583]	0.5714 [0.2857, 0.8571]	0.4819 [0.2597, 0.6875]	0.5380 [0.2924, 0.7758]	0.0065 [0.0064, 0.0066]
Weighted logistic regression	0.4348 [0.2902, 0.6111]	0.7143 [0.4982, 0.9286]	0.6329 [0.4167, 0.8025]	0.4006 [0.2339, 0.6969]	0.0054 [0.0045, 0.0064]
Always predict π (reference)	–	–	–	–	0.0067

Table 4: Odds ratios of the weighted logistic regression, per standard deviation (continuous) or flag present versus absent.

Feature	**Odds ratio [95% interval]**
$\tilde{d}_i$, log distance	0.368 [0.322, 0.436]
s_i, sine of month	1.161 [0.823, 1.404]
c_i, cosine of month	1.112 [0.893, 1.413]
ℓ_i, log lag	1.893 [1.384, 2.302]
k_i^{dead}	0.997 [0.938, 1.073]
k_i^{spec}	1.029 [0.953, 1.174]

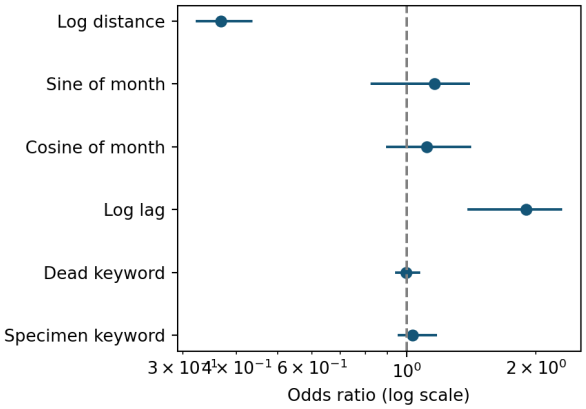


Figure 1: Odds ratios with 95% bootstrap intervals (log scale); the vertical line marks an odds ratio of 1.

Table 5: The ten highest-scoring unresolved reports. An asterisk marks use of the submission date.

Rank	GlobalID	Date	Distance (km)	Calibrated p
1	339CEB4E-EAC1-45C4-A8B5-51BD33A65854	2018-09-20	0.529	0.62365
2	9F458526-1FF1-4FEE-8592-CE767A4828CA	2010-04-20	7.828	0.54954
3	FA9E7423-0FA0-4D52-AE43-79F3A024608E	2014-04-24	6.995	0.49449
4	EE72574A-1F51-483C-8C76-6B097B8BCD84	2019-07-12	0.958	0.41619
5	EB4746A1-5A8B-4D7D-9149-D560988238EC	2019-10-15	0.871	0.37854
6	75E7FD91-2E03-4403-9E0B-443884012637	2019-07-23	2.314	0.29530
7	3D97022E-5F57-43BE-909B-A85326FB8E52	2011-08-20	14.155	0.27301
8	F766D086-08B9-40D4-9A28-DC57175E0503	2020-05-08	1.392	0.21897
9	E035D8C5-F345-4266-9EF3-3F29B0B42059	2018-07-20	7.271	0.21708
10	0D889239-8155-4F90-B1B2-712EA2C652EA	2019-08-10	5.335	0.20087

Table 6: Recall at capacity K (out-of-fold) and expected positives among the top K unverified reports.

K	Recall at K	Expected positives
10	6/14 [0.177, 0.711]	3.668
25	10/14 [0.419, 0.916]	5.793
50	11/14 [0.492, 0.953]	7.618
100	12/14 [0.572, 0.982]	9.388

Table 7: Average precision under relative distance error η (mean of 100 draws).

η	AP	Change
0	0.4006	0 (reference)
0.05	0.4001	−0.0005
0.10	0.3994	−0.0011
0.25	0.3955	−0.0051

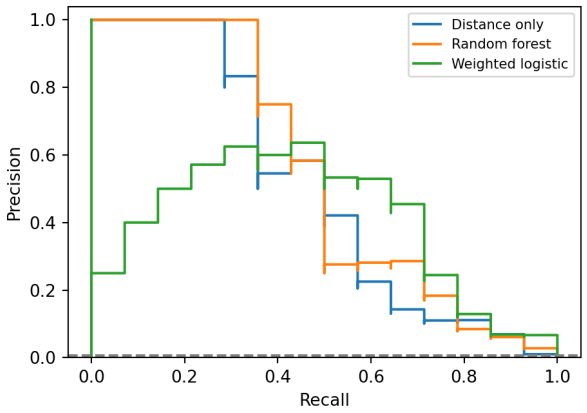


Figure 2: Out-of-fold precision–recall curves for the three models; the dashed line is the base rate $\pi = 0.0067$.

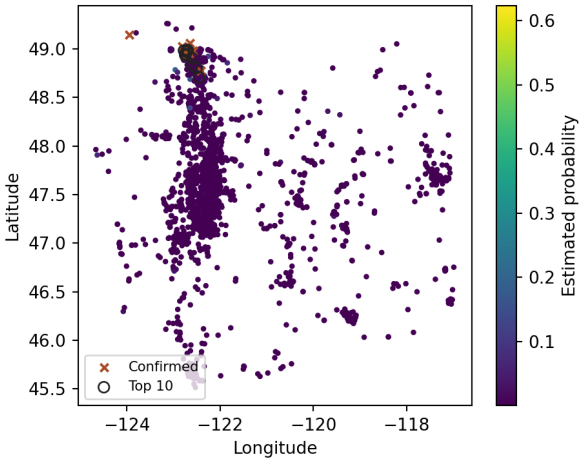


Figure 3: Unverified and unprocessed reports colored by p_i^{cal}, with the 14 positives marked and the reports of Table 5 circled.

5 Model II: Predicting the Spread

Dispersal kernel and prediction region. Each fall a new queen leaves her natal nest at $\mathbf{c}$, and the next spring she founds a nest at $\mathbf{c} + \mathbf{Z}$, where $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \sigma_D^2 I_2)$ in kilometers. The dispersal distance $D = \|\mathbf{Z}\|$ is then Rayleigh distributed, $P(D \le r) = 1 - \exp(-r^2/2\sigma_D^2)$. The problem statement gives a queen range of about 30 km [4], which we read as the distance within which 95% of queens settle. Setting $P(D \le R) = 0.95$ gives

$$\sigma_D = \frac{R}{\sqrt{-2\ln 0.05}} = \frac{30}{2.448} = 12.26 \text{ km}, \tag{9}$$

a mean dispersal distance of $\sigma_D\sqrt{\pi/2} = 15.4$ km. The kernel is isotropic, which is a simplification: it ignores the Strait of Georgia and Puget Sound, which shrink the habitable region, and human transport of queens in soil, nursery stock or cargo, which can carry one far beyond 30 km.

Positives from one nest are not independent draws (Assumption 5), so with n nest cluster locations $\mathbf{c}_1, \dots, \mathbf{c}_n$ in the current season, the predicted density of a new nest at $\mathbf{z}$ is

$$f(\mathbf{z}) = \frac{1}{n}\sum_{j=1}^{n} \frac{1}{2\pi\sigma_D^2} \exp\left(-\frac{\|\mathbf{z} - \mathbf{c}_j\|^2}{2\sigma_D^2}\right). \tag{10}$$

The 95% prediction region is the smallest area holding 95% of this density (Figure 4). If spread goes unchecked for t seasons, the displacements add and the 95% radius grows to $R\sqrt{t}$: 30 km after one season, $30\sqrt{2} = 42.4$ km after two.

Fit to the data and spread rate. With the 1 km rule, the four 2019 mainland positives form four clusters and the nine 2020 positives form five, because the five reports from 21

September to 1 October 2020 lie within 0.71 km of each other. The four 2019 mainland positives have centroid $(48.993°, -122.726°)$, near Blaine. The 7 June 2020 positive at $(48.7775°, -122.4186°)$ lies 32.8 km from it, just outside the 30 km radius. Under the kernel, $P(D > 32.8) = \exp(-32.8^2/(2 \cdot 12.26^2)) = e^{-3.58} = 0.028$, so one such distance is plausible. As goodness of fit, the share of 2020 clusters inside the 95% region built from the 2019 clusters is 4 of 5 (80%, exact 95% interval $[0.284, 0.995]$); five clusters give only a coarse check of coverage.

The 2019 positive near Nanaimo on Vancouver Island lies about 90 km from the Blaine centroid, across open water, with kernel probability $e^{-27} \approx 10^{-12}$. DNA evidence also showed the Washington and Vancouver Island hornets came from different nests [5], so we treat it as a separate introduction and exclude it. The region therefore had at least two independent introductions within a year, and no dispersal model can predict where a transported queen will appear next.

At report level, the centroid of the nine 2020 positives, $(48.960°, -122.595°)$, lies 10.2 km from the 2019 centroid; at cluster level the 2020 centroid is $(48.941°, -122.609°)$ and the shift is 10.3 km. We take this shift as the spread rate $\hat{v}$ (km per year), with 95% interval [2.6, 22.2] km per year from $B = 2000$ resamples of clusters within each year.

Answer to the first question. The spread can be predicted as a region, not as a point. Under the assumed kernel, each new nest has a 95% probability of lying within 30 km of its origin cluster: about $\pi \cdot 30^2 \approx 2{,}830$ km^2 around each cluster, less where circles overlap. After two unchecked seasons the radius is 42.4 km and the area about 5,650 km^2. Finer precision is not possible: the kernel rests on a handful of nest clusters, and people can carry a queen well outside the region, as at Nanaimo.

6 Model III: Updating the Model Over Time

Retraining triggers and schedule. The distance feature in Equation (3) depends on the set of confirmed positives, so one new positive changes the score of every report near it. Model I is retrained under three triggers:

1. **Within 24 hours of a new positive:** recompute d_i for every open report, refit, re-rank $\mathcal{U}$ and redraw the Model II region.
2. **Monthly from May to November, quarterly from December to April:** refit on the new labels and update the base rate below.
3. **On drift:** retrain if the Brier score of the M reports labeled in the past month exceeds $\mathrm{BS}_0 + 2s_B/\sqrt{M}$, where BS_0 is the cross-validated Brier score and s_B the standard deviation of its per-report terms $(p_i^{\mathrm{cal}} - y_i)^2$: $\mathrm{BS}_0 = 0.00542$, $s_B = 0.06412$, and the threshold at $M = 200$ is 0.01449.

New reports are scored on arrival, the ranked list is reissued weekly in season, and each spring before May α is re-tuned, Model II is refit on the new clusters and μ_L is recomputed.

Updating the base rate. Public awareness changes the share of positives, so the base rate gets a $\mathrm{Beta}(a_0, b_0)$ prior, updated with each batch of labels to $\pi \mid \text{data} \sim \mathrm{Beta}(a_0+N_1,\ b_0+N_0)$

with mean $\hat{\pi} = (a_0 + N_1)/(a_0 + b_0 + N_1 + N_0)$. With the uniform prior $a_0 = b_0 = 1$ the posterior is Beta$(15, 2070)$: $\hat{\pi} = 15/2085 = 0.0072$, 95% credible interval $[0.0040, 0.0112]$. When the estimate moves from π to π', the intercept of Model I shifts without a full refit:

$$\beta_0' = \beta_0 + \ln \frac{\pi'}{1-\pi'} - \ln \frac{\pi}{1-\pi}. \tag{11}$$

Worked example. If π rises from $14/2083 = 0.0067$ to 0.0100, then β_0 rises by $-4.595 - (-4.996) = 0.401$, so every report's odds are multiplied by $e^{0.401} = 1.49$, and the report of the calibration example after Equation (5) moves from $p^{\text{cal}} = 0.026$ to 0.039. The shift changes expected counts and the cost rule, never the ranking.

7 Model IV: Evidence of Eradication

While a population is present, positive clusters arrive as a Poisson process with rate μ per active month (Assumption 6), so zero positives over T active months has probability $e^{-\mu T}$; if the population is gone, zero is certain. With prior probability ρ_0 that the population is still present, Bayes' rule gives

$$P(\text{present} \mid 0 \text{ positives in } T) = \frac{\rho_0 e^{-\mu T}}{\rho_0 e^{-\mu T} + (1-\rho_0)}. \tag{12}$$

We declare eradication when this probability is at most $\gamma = 0.05$. Multiplying out the denominator and taking logarithms:

$$\rho_0 e^{-\mu T}(1-\gamma) \le \gamma(1-\rho_0) \iff \mu T \ge \ln \frac{(1-\gamma)\,\rho_0}{\gamma\,(1-\rho_0)} \iff T \ge T^* = \frac{1}{\mu} \ln \frac{(1-\gamma)\,\rho_0}{\gamma\,(1-\rho_0)}. \tag{13}$$

Since a smaller μ demands a longer quiet period, we use the lower 95% bound on μ. Let T_{obs} be the active months of 2020 in the data, May through October (the last submission is dated 23 October 2020), so $T_{\text{obs}} = 6$; 2019 is left out because detection effort differed. The mainland clusters first detected in these months number 2, 1, 0, 1, 1 and 0 (as reports: 2, 1, 0, 1, 4 and 1), so $k = 5$ and $\hat{\mu} = 5/6 = 0.83$ per active month. The exact Poisson lower bound is $\mu_L = \chi^2_{2k,\,0.025}/(2T_{\text{obs}})$, where $\chi^2_{2k,\,0.025}$ is the 2.5th percentile of the chi-squared distribution with $2k$ degrees of freedom: $\mu_L = 0.2706$ per active month and $T^* = 19.0$ active months at $\rho_0 = 0.9$.

Table 8 gives T^* for assumed values of μ, and Figure 5 shows the posterior falling with quiet months; the deployed value uses the estimated μ_L. The prior $\rho_0 = 0.5$ is neutral; $\rho_0 = 0.9$ is the cautious choice right after a nest is destroyed. For example, $\mu = 0.5$ and $\rho_0 = 0.5$ give $T^* = 2 \ln 19 = 5.9$ active months, close to one season.

Table 8: Quiet active months T^* needed to declare eradication at $\gamma = 0.05$

	$\mu = 0.25$	$\mu = 0.5$	$\mu = 1.0$
$\rho_0 = 0.5$ $(\ln 19 = 2.944)$	11.8	5.9	2.9
$\rho_0 = 0.9$ $(\ln 171 = 5.142)$	20.6	10.3	5.1

Checking the Poisson assumption. The variance-to-mean ratio of the monthly cluster counts is 0.68. A ratio well above 1 would mean bursts, making long quiet spells likelier while a population is present; we would then replace $e^{-\mu T}$ by the negative binomial zero probability, which lengthens T^*.

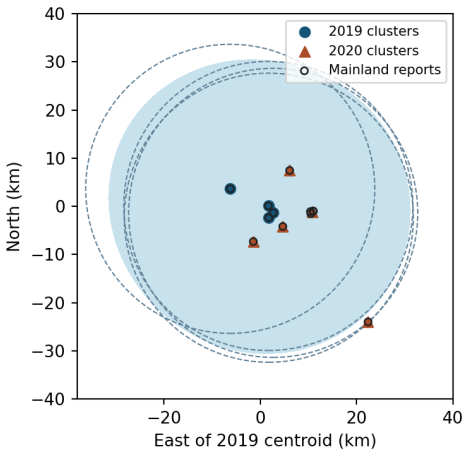


Figure 4: Mainland confirmed positives, 30 km circles around the 2019 nest clusters, and the 95% region of Equation (10) from the 2019 clusters.

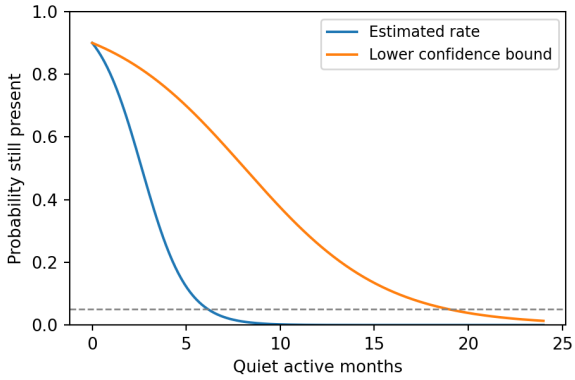


Figure 5: Equation (12) against quiet active months for $\mu = \hat{\mu}$ and $\mu = \mu_L$ at $\rho_0 = 0.9$. The dashed line is $\gamma = 0.05$; each curve crosses it at its T^*.

Decision rule. New queens overwinter in the soil and emerge the next spring [4], so one missed nest produces no reports until the following season. We therefore declare eradication only when (1) at least T^* active months pass with no confirmed positive, spanning *at least two consecutive active seasons*, and (2) every report that Model I ranked in the top K during that period was investigated and came back negative, so the silence comes from searching and finding nothing rather than from nobody looking.

8 Model Strengths & Limitations

8.1 Strengths

- **Leakage-safe validation:** the distance feature is recomputed inside every training fold, so each held-out positive is excluded from its own reference set. Reports from the same nest can appear in different folds, so these scores measure performance on held-out reports and may overstate performance on unseen nests.
- **Calibrated probabilities:** after the correction of Equation (5), the summed scores of the top K reports estimate how many hornets a week of visits will find.
- **Explicit error rate for eradication:** Model IV fixes $\gamma = 0.05$, uses the conservative μ_L and requires active searching, not just silence.

- **Cheap to maintain:** every component is closed form; a base-rate change needs only the intercept shift of Equation (11).

8.2 Limitations

- **Only 14 positives:** every interval is wide, and pooled recall moves in steps of 1/14.
- **Spread fit on a handful of nest clusters:** the coverage check cannot reject a moderately wrong kernel.
- **Reporter and search bias:** people report, and the state searches, near publicized sightings. This inflates the apparent power of the distance feature and lowers the score of a nest far from past positives.
- **Isotropic dispersal:** the kernel ignores water, terrain and human transport, so it misses long jumps such as the Nanaimo introduction.

9 Operational Deployment & Recommendations

1. **Work the ranked list weekly.** From May to November, investigate the K highest-scoring open reports (K = field capacity), plus any report above the cost cutoff C/L. At $K = 25$ the out-of-fold ranking recovered 10 of 14 confirmed sightings (exact 95% interval for recall $[0.419, 0.916]$).
2. **Close the loop and keep the schedule.** Label every visited report within the week; re-rank within 24 hours of a new positive; refit monthly in season, quarterly out of season, and revalidate each spring (Model III).
3. **Trap by the spread map.** Place traps inside the Model II 95% region, and widen to 42 km around any cluster whose nest was not found.
4. **Declare eradication only by rule:** at least 19.0 quiet active months (lower rate 0.2706 per active month), spanning at least two consecutive active seasons, with every top-K report searched and negative.
5. **Collect better data:** a mandatory photo field, phone GPS instead of geocoded addresses, and a log of traps that caught nothing.

References

[1] Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems (NeurIPS)*, 30, 4765–4774.

[2] GeeksforGeeks. Species distribution modeling in scikit-learn. https://www.geeksforgeeks.org/machine-learning/species-distribution-modeling-in-scikit-learn/.

[3] OpenAI. ChatGPT, model "GPT-6". Prompt: "Explain Hornet Project Modeling."

[4] COMAP. (2021). *2021 MCM Problem C: Confirming the Buzz about Hornets*. Mathematical Contest in Modeling.

[5] Penn State Extension. (2020). *Asian Giant Hornets* (article, updated May 6, 2020; no author named). Supplied as `2021MCM_ProblemC_Vespamandarina.pdf` with the 2021 MCM Problem C data.

A Python Code: Classification, Spread and Eradication Models

Run `model.py`, `spread.py`, and `fill.py` from a directory containing the extracted contest data and the original LaTeX draft at `upload/Pasted text.txt`. The scripts use NumPy, pandas, SciPy, scikit-learn, Matplotlib, and openpyxl.

B Python Data Cleaning Code

```
import pandas as pd
import numpy as np

dataset = pd.read_excel("2021MCMProblemC_DataSet.xlsx")

print("Dataset shape:", dataset.shape)
print("\nMissing values:")
print(dataset.isna().sum())

print("\nDuplicate records:")
print(dataset["GlobalID"].duplicated().sum())

dataset["Detection Date"] = pd.to_datetime(
dataset["Detection Date"], errors="coerce")
dataset["Submission Date"] = pd.to_datetime(
dataset["Submission Date"], errors="coerce")
invalid_dates = (dataset["Detection Date"] > dataset["Submission Date"])

print("\nInvalid dates:", invalid_dates.sum(), "\n")

dataset["target"] = dataset["Lab Status"].map({
    "Positive ID": 1,
    "Negative ID": 0
    })

print("Classification counts:")
print(dataset["target"].value_counts(dropna=False))

invalid_locations = (
    ~dataset["Latitude"].between(-90, 90)
    | ~dataset["Longitude"].between(-180, 180)
)

print("\nInvalid geographic coordinates:")
print(invalid_locations.sum())
```

```
outside_region = (
    ~dataset["Latitude"].between(45.5, 49.1)
    | ~dataset["Longitude"].between(-125.0, -116.8)
)

print("\nLocations outside study region:")
print(outside_region.sum())

print("\nSuspicious location records:")
print(dataset.loc[
    invalid_locations | outside_region,
    ["GlobalID", "Latitude", "Longitude"]
].head(20))
```